DISTRIBUTIVE LATTICE POLYMORPHISM ON REFLEXIVE GRAPHS

MARK SIGGERS

Abstract. In this paper we give two characterisations of the class of reflexive graphs admitting *distributive lattice polymorphisms* and use these characterisations to address the problem of recognition: for a reflexive graph G in which no two vertices have the same neighbourhood, we find a polynomial time algorithm to decide if G admits a distributive lattice polymorphism.

1. Introduction

1.1. **Motivation.** It is well known (see for example [3]) that Constraint Satisfactions Problems, which provide a formulation for many combinatorial problems, can be stated as the problem of finding a homomorphism between structures. Moreover ([8]) any such homomorphism problem can be reduced to the retraction problem for *reflexive* graphs- graphs in which every vertex has a loop.

The problem $\mathrm{Ret}(G)$ of retraction to G is NP-complete for most reflexive graphs G and in the case that the problem is known to be polynomial time solvable for some G, the polynomial time algorithm is tied to the existence of a *polymorphism* on G– an operation $f : V(G)^d \to V(G)$ that preserves edges– satisfying some nice identity.

A reflexive graph G is a *lattice* graph if it has a *compatible lattice*, a lattice on its vertex set such that the meet and join operations, $\wedge$ and $\vee$, are polymorphisms of G. It is a *distributive lattice graph* or DL-*graph* if the lattice L is distributive; we call (G, L) a DL-*pair*.

It was shown in [4] that $\mathrm{Ret}(G)$ can be solved, for a structure G, by a linear monadic Datalog program with at most one extensional predicate per rule, (i.e., G has caterpillar duality), if and only if it is a retract of a DL-graph.

To give a bit more context, a d-ary polymorphism $f : G^d \to G$ is a *totally symmetric idempotent* (TSI) polymorphism if $f(v, v, \ldots, v) = v$ for all $v \in V(G)$ and if $f(v_1, \ldots, v_d) = f(u_1, \ldots, u_d)$ whenever $\{v_1, \ldots, v_d\} = \{u_1, \ldots, u_d\}$. The class TSI of reflexive graphs G admitting TSI polymorphisms of all aritites is important, as this is the class for which $\mathrm{Ret}(G)$ can be solved by a monadic Datalog program with at most one extensional predicate per rule, (i.e., G has tree duality).

It is of interest to get a graph theoretic characterisation of the class TSI. The two main sources of TSI polymorphisms, are near-unanimity (NU) polymorphisms, and semilattice (SL) polymorphisms. While NU polymorphisms have been well studied, and the classes of graphs admitting them have several nice characterisations, no such

Date: August 18, 2016.

Key words and phrases. Lattice Polymorphism, CSP, Distributive Lattice, Reflexive Graph, Recognition.

Supported by the National Research Foundation (NRF) of Korea (2014-06060000).

study had been attempted for SL polymorphisms until [14], where we looked at the family of reflexive graphs admitting SL polymorphisms.

A (meet) semilattice ordering on the vertices of a graph defines a 2-ary operation $\vee$ on the vertex set. If this operation is a polymorphism of the graph then it is an SL *polymorphism* of the graph. The problem of characterising reflexive graphs admitting SL polymorphsims was difficult, and we restricted our attention to those graphs G that admit SL polymorphisms for which the Hasse diagram of the semilattice ordering is a tree, and a subgraph of G. We showed that the class of such graphs extends the class of chordal graphs. We were unable to say much in the case that the semilattice was not a tree. The other extreme is when the ordering is a lattice– the Hasse diagram is not a tree except in the trivial case that the lattice it is a chain. This leads us to consider lattice polymorphisms.

1.2. **Results.** In this paper we give two explicit characterisations of the class of reflexive DL-graphs, and use these characterisations to address the problem of recognition.

For our first characterisation, we recall a well known result of Birkhoff [2]. For a poset P, a subset D is a *downset* if $b \in D$ and $a \leq b$ implies $a \in D$. The family $\mathcal{D}(P)$ of all downsets of P is a distributive lattice under the ordering $\subseteq$. The meet and join operations are $\cap$ and $\cup$, respectively. Birkhoff showed that for any distributive lattice L, L is isomorphic to $\mathcal{D}(J_L)$ for a unique poset J_L, (the poset of join irreducible elements of L).

Viewing a comparability $a \leq b$ as an arc (a, b), a poset P is just a transitive acyclic (except for loops) reflexive digraph. So we can talk of a sub-digraph A of P.

Definition 1.1 ($G(P, A)$)**.** For a poset P and a sub-digraph A of P, let $G = G(P, A)$ be the graph on $\mathcal{D}(P)$, in which two downsets $D, D' \in \mathcal{D}(P)$ are adjacent if A contains all arcs (x, y) of P for which x and y are in either $D - D'$ or $D' - D$.

See Figure 1 for an example. The left side shows a poset P represented by its Hasse diagram (defined in Section 2) in thick light edges, and a sub-digraph A in dark edges, missing only the arc (b, c). On the right is the downset lattice $\mathcal{D}(P)$ again represented by its Hasse diagram, and the graph $G(P, A)$ missing only edges between vertices one of which contains b and c and the other of which contains neither of them.

Our first main theorem, Theorem 4.3 says that for any DL-pair (G, L), G is isomorphic to $G(J_L, A)$ for some sub-digraph A of J_L. Showing, in Lemma 4.4 that $G(P, A)$ is always a DL-graph, we get the following characterisation of reflexive DL-graphs.

Corollary 1.2. *A reflexive graph G is a* DL*-graph if and only if there is there is a poset P and a sub-digraph A such that $G \cong G(P, A)$.*

In [11] it was shown that a graph (without loops) is a *proper interval graph* if and only if there is an ordering of its vertices such that if it satisfies the so-called *min-max* identity:

$$(u' \leq u \leq v \leq v' \text{ and } u' \sim v') \Rightarrow u \sim v \tag{1}$$

We take this as our definition of a proper interval graph in the reflexive context, and say a proper interval graph is in *min-max form* if its vertices are labelled $\{0, 1, \ldots, n\}$ for some n so that it satisfies (1).

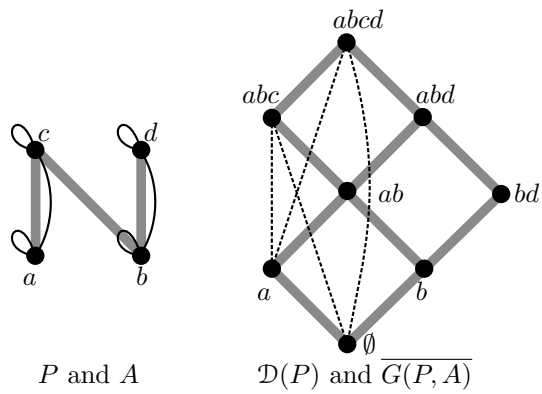


FIGURE 1. Poset P and lattice $\mathcal{D}(P)$ in thick light edges. Digraph A and (the complement of) graph $G(P, A)$ in dark.

Simple arguments (see Fact 2.1) show that any reflexive graph that is compatible with a chain lattice is a proper interval graph. As any distributive lattice is embeddable in a product of chains, it follows that any DL-graph is a subgraph of a categorical product (defined in Section 2) of proper interval graphs. In fact Dilworth [6] showed, and we recall this in more detail in Section 5, that any chain decomposition of P yields an embedding of $\mathcal{D}(P)$ into a product of chains. The embedding shown in Figure 1 comes from the decomposition of P into the chains $a \prec c$ and $b \prec d$. Figure 2 shows the embedding corresponding to the decomposition of P into the chains a, $b \prec c$, and d. For the embedding in Figure 2 the graph $G(P, A)$ is an induced subgraph of proper interval graphs, in this case paths, on the chain factors. It turns out that this happens when certain edges in the complement of A in P are contained in the chain decomposition of P.

For any DL-pair (G, L), we get, in Theorem 5.1, an embedding of L into a product of chains such that G is an induced subgraph of $\mathcal{G}$. Further, it is an induced subgraph of quite a particular form.

The vertex set of a product $\mathcal{G} = \prod_{i=1}^{d} G_i$ of proper interval graphs G_i is a set of d-tuples $x = (x_1, \ldots, x_d) \in \prod_{i=1}^{d} \{0, 1, \ldots, n_i\}$ for some $n_1, \ldots, n_d$. A *vertex interval* of $\mathcal{G}$ is the set

$$[\alpha_{[i]}, \beta^{[j]}] = \{x \in V(\mathcal{G}) \mid \alpha \leq x_i \text{ and } x_j \leq \beta\}$$

for some $i, j \in [d]$, $\alpha \leq n_i$ and $\beta \leq n_j$. (See right side of Figure 2.)

Our following, second, characterisation of reflexive DL-graphs is immediate from Theorem 5.1.

Corollary 1.3. *A reflexive graph G is a* DL*-graph if and only if it is the induced subgraph of a product $\mathcal{G} = \prod_{i=1}^{d} G_i$ of proper interval graphs G_i in min-max form, that we get by removing vertex intervals.*

A reflexive graph G is R-thin if no two vertices have the same neighbourhood. For questions of $\mathrm{Ret}(G)$, one may always assume that G is R-thin as there are simple linear time reductions between $\mathrm{Ret}(G)$ and $\mathrm{Ret}(G^R)$ where G^R, defined formally in Section 6.4, is the R-thin graph we get from G by removing all but one vertex

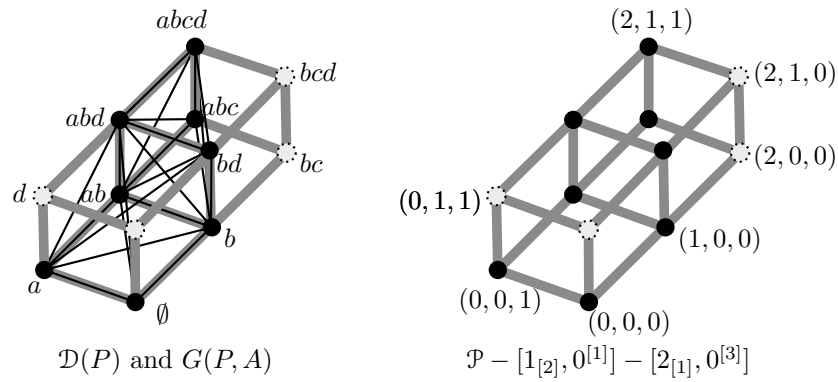


FIGURE 2. Left: The lattice $\mathcal{D}(P)$ from Figure 1 embedded in a product of three chains, and the graph $G(P, A)$ from Figure 1 embedded as an induced subgraph of the product of paths on those chains. Right: The usual labelling on the product of chains showing $\mathcal{D}(P)$ as $\mathcal{P} - [1_{[2]}, 0^{[1]}] - [2_{[1]}, 0^{[3]}]$.

from every set of vertices sharing the same neighbourhood. In Section 6 we prove the following.

Theorem 1.4. *There is a polynomial time algorithm to decide whether or not an R-thin reflexive graph is a* DL*-graph.*

In fact, we give a polynomial time algorithm that not only decides if a given R-thin graph has a compatible distributive lattice, but if it does, finds one (actually all) such lattices.

While for questions about $\mathrm{Ret}(G)$ one may assume that G is R-thin, distributive lattice polymorphisms are unusual in the fact that the existence of a compatible distributive lattice for G^R does not imply the existence of one for G. We finish of Section 6 with some notes about deciding if a non R-thin graph is a DL-graph.

2. Definitions, Notation, and Basic Observations

For any element u of any ordering, $\langle u]$ is the set of elements below u, and $[u\rangle$ is the set of elements above it. We write $a \prec b$ if b *covers* a; that is, if $a < b$ and there is no x such that $a < x < b$. It is standard to depict a poset by it's Hasse diagram– its sub-digraph of covers– and to depict direction of the covers simply by assuming that the greater element is higher on the page. As we draw a lattice and a graph on the same set of vertices, the edges of our Hasse diagram are the thicker lighter edges, and the graph edges are thin and dark.

Recall that a *lattice* L is a partial ordering on a set such that the greatest lower bound and least upper bound are uniquely defined for any pair of elements. These define the meet, $\wedge$, and join, $\vee$, operations respectively. It is a basic fact that the operations $\wedge$ and $\vee$ and the lattice defined each other by the identities

$$u \leq v \iff (u \wedge v) = u \text{ and } u \leq v \iff (u \vee v) = v.$$

The lattice is *distributive* if the meet and join distribute. As our lattices are finite the meet and join operations are well defined for any set of elements and there is a maximum element, or *unit*, denoted $\mathbf{1}$, and a minimum element or *zero* denoted $\mathbf{0}$.

A lattice L on the vertices of a graph G was defined to be compatible if its meet and join operations $\wedge$ and $\vee$ are polymorphisms. Explicitly, L is compatible with G if and only if the following holds, where '$\sim$' denotes adjacency in G:

$$(u \sim u' \text{ and } v \sim v') \Rightarrow (u \wedge v \sim u' \wedge v' \text{ and } u \vee v \sim u' \vee v'). \tag{2}$$

Along with (1), there is another useful property of an ordering of vertices.

$$u \sim v \sim w \text{ and } (u \leq v \geq w \text{ or } u \geq v \leq w) \Rightarrow u \sim w \tag{3}$$

Fact 2.1. *For a compatible lattice ordering of a reflexive graph G identities* (1) *and* (3) *hold for all vertices u, v, w of G.*

Proof. For (1), as $v \sim v$ we get $u = u \wedge v \sim w \wedge v = v$ and $v = u \vee v \sim w \vee v = w$, as needed. For (3), assuming that $u \leq v \geq w$, we get $u = (u \wedge v) \sim (v \wedge w) = w$. □

The product $L_1 \times L_2$ of two lattices is the ordering on the set $L_1 \times L_2$ defined by

$$(a_1, a_2) \leq (b_1, b_2) \text{ if } a_i \leq_i b_i \text{ for } i = 1, 2,$$

and the operations $\vee$ and $\wedge$ of the product are defined componentwise from the corresponding operations of the factors. Thus the product of distributive lattices is a distributive lattice. The *(categorical) product* of two graphs G_1 and G_2, is the graph $G = G_1 \times G_2$ with vertex set $V(G_1) \times V(G_2)$ and edgeset

$$\{(u_1, u_2)(v_1, v_2) \mid u_i v_i \in G_i \text{ for } i = 1, 2\}.$$

The following is standard.

Lemma 2.2. *If the reflexive graph G_i is compatible with the lattice L_i for $i = 1, 2$ then $G_1 \times G_2$ is compatible with $L_1 \times L_2$.*

Proof. Let $(u_1, u_2) \sim (v_1, v_2)$ and $(u_1', u_2') \sim (v_1', v_2')$ in $G_1 \times G_2$. Then

$$(u_1, u_2) \wedge (u_1', u_2') = (u_1 \wedge u_1', u_2 \wedge u_2') \sim (v_1 \wedge v_1', v_2 \wedge v_2') = (v_1, v_2) \wedge (v_1', v_2'),$$

and similarly $(u_1, u_2) \vee (u_1', u_2') = (v_1, v_2) \vee (v_1', v_2')$. □

A sublattice L' of a lattice L is any subset that is closed under the meet and join operations. The following is clear from the definition of compatibility.

Fact 2.3. *If a graph G is compatible with a lattice L, and L' is a sublattice of L, then the subgraph G' of G induced by L' is compatible with L'.*

A *conservative set* (or subalgebra) in a reflexive graph G is an subset $S \subset V(G)$ that is the intersection of sets of the form $\{x \in V(G) \mid d(x, x_0) \leq d\}$ for some vertex x_0 and integer d. Components and maximal cliques are examples of conservative sets. It is a basic fact, (see [3]), that a conservative set of a graph is closed under any polymorphism. We use this to prove the following, which allows us to restrict our attention to connected graphs.

Lemma 2.4. *A graph is a (distributive) lattice graph if and only if each component is.*

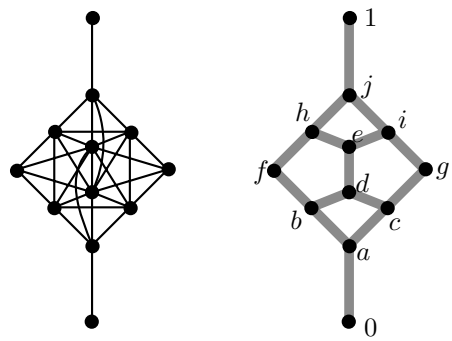


FIGURE 3. Graph (left) with compatible lattice (right) but no compatible distributive lattice

Proof. If a graph is disconnected, and each of its components has a compatible lattice L_i, then let L be the *simple join* of the component lattices; that is, let L be the lattice on the set $\bigcup_{i=1}^{d} L_i$ with the ordering defined by $x \leq y$ if $x \leq y$ in some L_i or if $x \in L_i$ and $y \in L_j$ for $i < j$. It is easy to check that this lattice is compatible with G, and that it is distributive if the component lattices are.

On the other hand, if a disconnected graph has a compatible lattice, then as each component is a subalgebra, and subalgebras are closed under polymorphisms, each component is closed under the lattice operations. Thus each component induces a sublattice, so is compatible with the component by Fact 2.3. If a lattice is distributive, then so is any sublattice. □

The following, which does not hold for semilattices, is a huge simplifaction.

Proposition 2.5. *For a connected reflexive graph G with a compatible lattice L, the Hasse diagram of L is a subgraph of G.*

Proof. It is enough to show for any cover $v \prec u$, that uv is an edge of G.

Observe first that the upset $[v\rangle$ is a connected subgraph of G. Indeed as G is connected, for u_0 and u_p in $[v\rangle$, there is a path $u_0 \sim u_1 \sim \cdots \sim u_p$ between them in G. So $(v \vee u_0) \sim (v \vee u_1) \sim \cdots \sim (v \vee u_p)$ is a walk between them in $[v\rangle$.

The same proof in connected $[v\rangle$ then shows that the downset $\langle u]$ in $[v\rangle$ is connected. But it contains only u and v, so uv is an edge. □

3. Some Examples

As all but the minimum and maximum vertex of a lattice must have at least one cover and be covered by one other vertex, the following is immediate from Proposition 2.5.

Example 3.1. For a connected reflexive graph G with a degree one vertex v, v must be the minimum or maximum vertex of any compatible lattice L. In particular, the only reflexive trees with compatible lattices are paths.

Proposition 3.2. *Neither the class of graphs admitting compatible lattices, nor the class admitting compatible distributive lattices, are closed under retraction.*

Proof. It is easy to see that the reflexive biclique $K_{1,4}$ is a retract of the product P_2^2 of two reflexive paths. P_2^2 has a distributive lattice by Lemma 2.2, but $K_{1,4}$ does not, by Example 3.1. □

Proposition 3.3. *There are graphs that have compatible lattices but have no compatible distributive lattices.*

Proof. Let G be the graph on the left of Figure 3. It is easy but tedious to verify, using (2) that the non-distributive lattice shown on the right is compatible. We show that there is no distributive lattice that is compatible with G.

Assume, towards contradiction, that G has a compatible distributive lattice. By Proposition 2.5, $\mathbf{0}$ and $\mathbf{1}$ must be the vertices labelled 0 and 1 in the figure. Further $\mathbf{0}$ must have unique cover a and $\mathbf{1}$ must cover j. So $\langle j] \cap [a\rangle$ is a distributive sublattice with zero a and unit j.

As the set $\{d, e\}$ is the intersection of maximal cliques, it is a conservative set, so induces a sublattice. The only 2 element lattices is the chain, so we may assume, without loss of generality, that $d \leq e$.

The set $\{d, e, h\}$ is also an intersection of maximal cliques, so induces a sublattice of three elements, so must also be a chain. If $h \leq e$ then by (3) a and h are adjacent, so $h \geq e$. Similarly $b \leq d$.

The set $\{b, d, e, f, h\}$ is a maximal clique, so induces a lattice. As b is not adjacent to a or j, it follows from (3) that it can neither be above or below d or e, so it is incomparable with them. Thus the sublattice induced on $\{b, d, e, f, h\}$ is as shown in the figure. It is well known that no lattice with this lattice as a sublattice is distributive. □

4. Downset Construction

Our main result of this section is Theorem 4.3. Before we prove it, we make some easy observations about the construction $G(P, A)$ of Definition 1.1. Recall that A is a sub-digraph of a poset P; it will always have the same vertex set as P We write $a \to b$ to mean that (a, b) is an arc of A.

An arc (x', y') of P is *useless* in a sub-digraph A if there is $(x, y) \notin A$ with $x' \leq x \leq y \leq y'$ (and $x' \neq x$ or $y' \neq y$). Removing all useless arcs from A it clearly satisfies the following directed version of (1)

$$(u' \leq u \leq v \leq v' \text{ and } u' \to v') \Rightarrow u \to v \tag{4}$$

Lemma 4.1. *The graph $G(P, A)$ is unchanged by adding or removing useless arcs from A. Thus A may be assumed to satisfy* (4).

Proof. Let $x' \leq x \leq y \leq y'$, and $x \not\to y$ in A. For any two downsets D and D' of P with x', y' in $D - D'$, we clearly have that x, y are in $D - D'$ as well, and so $D \not\sim D'$ in $G(P, A)$ whether $x' \to y'$ or not. □

For a sub-digraph A of a poset P, let A^{c} be the sub-digraph of P with arc set

$$\{(x, y) \mid (x, y) \in P - A\}.$$

The following is a useful alternate definition of adjacency in $G(P, A)$.

Lemma 4.2. *Where A is a sub-digraph of a poset P, and D and D' are in $\mathcal{D}(P)$, D and D' are adjacent in $G(P, A)$ if and only if the following hold for all vertices x and y.*

- *If $x \in D$ and $(y,x) \in A^{\mathrm{c}}$ then $y \in D'$, and*
- *If $x \in D'$ and $(y,x) \in A^{\mathrm{c}}$ then $y \in D$.*

Proof. The definition of adjacency of D and D' is clearly equivalent to the statement that neither of $D - D'$ or $D' - D$ induce an edge of A^{c}. That $D - D'$ induces no edge in A^{c} is equivalent to the statement that for all (y,x) in A^{c} with x, y in D, either $y \notin D'$ or $x \notin D'$. As D and D' are downsets, this reduces to the statement that for all (y,x) in A^{c} with $x \in D$, $y \notin D'$. □

Theorem 4.3. *For any reflexive graph G compatible with a distributive lattice L, $G \cong G(J_L, A)$ for a unique sub-digraph A of J_L satisfying* (1).

Proof. Let G be compatible with a distributive lattice L. By [2] we have that $L \cong \mathcal{D}(J_L)$, so we denote vertices of G by downsets of the poset J_L.

We define a sub-digraph $A = A(G, J_L)$ of J_L as follows. For a vertex p of J_L, let

$$C_p = X - [p\rangle = \bigcup\{X \in \mathcal{D} \mid p \notin X\}$$

be the maximum downset not containing p. For each arc $y \to x$ of J_L, let $y \to x$ be in A if $\langle x] \cap C_y$ and $\langle x]$ are adjacent in G. We show that $G \cong G(J_L, A)$.

Let D and D' be adjacent downsets of J_L. To show that D and D' are adjacent in $G(J_L, A)$, it is enough to show, without loss of generality, that any arc $(y,x) \in J_L$ for $x, y \in (D - D')$, is in A. So we must show that $\langle x] \cap C_y$ and $\langle x]$ are adjacent. As $D' \sim D$ and $\langle x] \sim \langle x]$ in G we have that

$$(\langle x] \cap D') \sim (\langle x] \cap D) = \langle x].$$

But as $y \notin D'$ we have that $D' \leq C_y$, and so we also have

$$\langle x] \cap D' \leq \langle x] \cap C_y \leq \langle x];$$

and so by (1) we get $\langle x] \sim (\langle x] \cap C_y)$, as needed.

On the other hand, let D and D' be non-adjacent downsets of J_L. Then we must show that there is some arc $(y,x) \in J_L$ for $x, y \in (D - D')$ or $x, y \in (D' - D)$ that is not in A. Assume that all such arcs with $x, y \in (D - D')$ are in A. Then for each, we saw above that $\langle x] \sim (\langle x] \cap C_y)$. Fixing x and taking the intersection over all $y \leq x$ in $D - D'$, we get

$$\langle x] = \bigcap \langle x] \sim \bigcap (\langle x] \cap C_y) = \bigcap (\langle x] - [y\rangle) =: T_x$$

where $T_x = \langle x] - \bigcup [y\rangle$ is contained in $\langle x] - D'$ as the union is over all $y \leq x$ that are in $D - D'$. Now taking the union over all $x \in D - D'$ we get that

$$D = \bigcup \langle x] \sim \bigcup T_x \subseteq \bigcup (\langle x] - D') \subseteq (D - D').$$

By (1) we get that $D \sim D - D'$.

Similarily we get that $D' \sim (D \cap D')$. But as $D \cap D' \leq D, D'$ we get from (3) that $D \sim D'$, a contradiction.

Now, Lemma 4.1 allows us to assume that A satisfies (1). The uniqueness of A then follows by observing that $G(J_L, A')$ would be different for any other sub-digraph A' of J_L satisfying (1): this is simple let (y,x) be an arc of A' but not A. Then the edge $\langle x] \sim \langle y]$ is in $G(J_L, A')$ but not in $G(J_L, A)$. □

Corollary 1.2 is immediate from Theorem 4.3 and the following lemma.

Lemma 4.4. *If P is a poset and $A \subseteq P$, then $\mathcal{D}(P)$ is compatible with $G = G(P, A)$.*

Proof. We use Lemma 4.2 for the definition of adjacency in G. Assume that $D \sim D'$ and $E \sim E'$. We must show that $(D \cup E) \sim (D' \cup E')$ and $(D \cap E) \sim (D' \cap E')$. For the former, let $x \in D \cup E$ and (x, y) be in A^{c}. Then $x \in D$ or E, so as $D \sim D'$ and $E \sim E'$, we have that $y \in D'$ or E'. Thus $y \in D' \cup E'$. That $x \in D' \cup E'$ implies $y \in D \cup E$ is the same, so $(D \cup E) \sim (D' \cup E')$. The proof of the latter is similar. □

Now, consider $G(C, A)$ where C is a chain. All downsets are of the form $\langle c]$ for some $c \in C$, or $\emptyset =: \langle -1]$. As we may assume that A satisfies (1), two downsets $\langle x]$ and $\langle y]$, for $y \leq x$ are adjacent if and only if $(y + 1, x)$ is an arc of A. The following is then clear, and is the starting point of our next characterisation of reflexive DL-graphs.

Fact 4.5. *For a chain C and a sub-digraph A, $G(C, A)$ satisfies* (1)*, so is a proper interval graph.*

Proof. Assume that $\langle u] \subsetneq \langle v] \subsetneq \langle w]$ and $\langle u] \sim \langle w]$ in $G(C, A)$. So $u < v < w$ and $u + 1 \to w$ in A. As we may assume that A satisfies (1) we have $u + 1 \to v$ and $v + 1 \to w$ in A, and so $\langle u] \sim \langle v]$ and $\langle v] \sim \langle w]$. □

5. Reflexive DL-graphs as subgraphs of products of Proper Interval Graphs

In this section we prove Theorem 5.1 and give some related results.

Theorem 5.1. *For any* DL*-pair (G, L), there is an embedding $L \cong \mathcal{P} - \mathcal{V}$, where $\mathcal{V}$ is a union of intervals of the form $[\alpha_{[i]}, \beta^{[j]}]$, of L into a product $\mathcal{P} = \prod P_i$ of chains. Further, for each P_i there is a proper interval graph G_i compatible with P_i such that G is the subgraph of $\mathcal{G} = \prod_{i=1}^{d} G_i$ induced by vertices in L.*

Before proving this, we observe that this gives Corollary 1.3.

Proof of Corollary 1.3. If G is a DL-graph then Theorem 5.1 gives us the necessary embedding of G into a product of proper interval graphs.

On the other hand, assume we get G from a product $\mathcal{G} = \prod G_i$ of proper interval graphs G_i in min-max form by removing vertex intervals $[\alpha_{[i]}, \beta^{[j]}]$. The ordering on the G_i is a chain lattice P_i, so induces on $V(\mathcal{G})$ a lattice $\mathcal{P} = \prod P_i$. By a result in [17], the subset induced by removing sets of the form $[\alpha_{[i]}, \beta^{[j]}]$ is a sublattice. By Fact 2.3 it is therefore compatible with the subgraph G of $\mathcal{G}$ that it induces. □

We start with some results from the literature that will help us prove Theorem 5.1.

5.1. **Setup for the proof of Theorem 5.1.** The first statement of Theorem 5.1 is acually from [18]. We explain, as we will have to build on this. Generalising the notion of a *chain decomposition* in which the subchains must be disjoint, a *chain cover* of a poset P is family $\mathcal{C} = \{C_1, \ldots, C_d\}$ of subchains of P such that every element of P is in one chain. Given $\mathcal{C}$, label the elements of P so that the subchain C_i is $1^{(i)} \prec \cdots \prec n_i{}^{(i)}$ for some n_i; if an element is in more than one chain, it gets more than one label. It is clear that a downset D of P is uniquely defined by the tuple $e_{\mathcal{C}}(D) = (x_1, \ldots, x_d)$ where $x_i = |D \cap C_i|$. (Note that D is thus the downset generated by the set $\{x_i{}^{(i)} \mid i \in [d]\}$.)

As Dilworth [6] observed in the case that $\mathcal{C}$ is a decomposition, we observed in [18] that $e_{\mathcal{C}}$ is in fact a lattice embedding of $\mathcal{D}(P)$ into the product of chains $\mathcal{P}_{\mathcal{C}} = \prod_{i=1}^{d} P_i$ where P_i is the chain $0 \prec 1 \prec \cdots \prec n_i$ with one more element than C_i. Thus by Birkoff's result from [2], every chain cover of J_L gives an embedding $e_{\mathcal{C}}$ of L as a sublattice of a product $\mathcal{P}_{\mathcal{C}}$ of chains.

In (Corollary 6.6 of) [18] we showed that every embedding of L as a sublattice of a product of chains such that L contains the zero and unit of the product, is $e_{\mathcal{C}}$ for some chain cover $\mathcal{C}$ of J_L.

The following notation will also be useful, and explains the notation $[\alpha_{[i]}, \beta^{[j]}]$. Given a product of chains $\mathcal{P}_{\mathcal{C}} = \prod_{i=1}^{d} P_i$, let

$$\alpha_{[i]} = (\underbrace{0, \ldots, 0, \alpha}_{i}, 0, \ldots, 0) \qquad \text{and} \qquad \beta^{[j]} = (n_1, \ldots, n_{j-1}, \beta, n_{j+1}, \ldots, n_d)$$

for all $i, j \in [d]$ and α, β with $0 \leq \alpha \leq n_i$ and $0 \leq \beta \leq n_j$.

5.2. **Proof of Theorem 5.1.** Let (G, L) be a DL-pair, so by Theorem 4.3, $G = G(J_L, A)$ for some sub-digraph A of J_L, and let $\mathcal{C}$ be a chain cover of J_L. For each chain $C_i \in \mathcal{C}$ let A_i be the subgraph of A induced by the vertices of C_i. By Fact 4.5 we have that $G_i = G(C_i, A_i)$, on the chain P_i is a proper interval graph. Let $\mathcal{G} = \prod G_i$ be the product of these proper interval graphs. The embedding $e_{\mathcal{C}} : \mathcal{D}(J_L) \to \mathcal{P}_{\mathcal{C}}$ embeds $V(G)$ as a subset of $V(\mathcal{G})$.

In [18], we observed the following, using a result of [17].

Proposition 5.2. *Where L is the image of the embedding $e_{\mathcal{C}} : \mathcal{D}(J_L) \to \mathcal{P}_{\mathcal{C}}$,*

$$L = \mathcal{P}_{\mathcal{C}} - \mathcal{V}$$

where $\mathcal{V}$ is the union, over comparable pairs $(\beta + 1)^{(j)} \leq \alpha^{(i)}$ in J_L, of the intervals

$$[\alpha_{[i]}, \beta^{[j]}] = \{x \in \mathcal{P}_{\mathcal{C}} \mid \alpha \leq x_i \text{ and } x_j \leq \beta\}.$$

So any chain cover $\mathcal{C}$ of J_L yields an embedding of L into $\mathcal{P}_{\mathcal{C}}$ as required by the first statement of Theorem 5.1. To show that G can be taken as an induced subgraph of $\mathcal{G}$ we need a similar statement about how the edges of G relate to those of $\mathcal{G}$. We get this by looking at how our construction $G(J_L, A)$ behaves notationally under the embedding $e_{\mathcal{C}}$.

To finish proving Theorem 5.1 we must show that $e_{\mathcal{C}}$ embeds G as an induced subgraph of $\mathcal{G}$. This is not true of every chain cover $\mathcal{C}$, but every chain cover embeds it as a subgraph.

For each $i, j \in [d]$ and $0 \leq \alpha \leq n_i$ and $0 \leq \beta \leq n_j$, let

$$[\alpha_{[i]}\rangle \times \langle\beta^{[j]}] := \{\{x, y\} \mid \alpha \leq x_i \text{ and } y_j \leq \beta\}.$$

Lemma 5.3. *Where $\mathcal{C}$ is a chain cover of J_L, $G = G(J_L, A)$ is a subgraph of the product $\mathcal{G} = \prod G_i$ of proper interval graphs $G_i = (C_i, A_i)$. In fact $G = \mathcal{G} - \mathcal{V} - \mathcal{E}$ where*

$$\mathcal{V} = \bigcup \{[\alpha_{[i]}, \beta^{[j]}] \mid (\beta + 1)^{(j)} \leq \alpha^{(i)} \in J_L\}$$
$$\mathcal{E} = \bigcup \{[\alpha_{[i]}\rangle \times \langle\beta^{[j]}] \mid (\beta + 1)^{(j)} \to \alpha^{(i)} \in A^{\mathrm{c}}\}.$$

Proof. Let D_x and D_y be downsets of J_L and let $x = e_{\mathcal{C}}(D_x)$ and $y = e_{\mathcal{C}}(D_y)$.

If $D_x \sim D_y$ we have in particular that for each i, $x_i + 1^{(i)} \to {y_i}^{(i)}$ if $x_i < y_i$ and $y_i + 1^{(i)} \to {x_i}^{(i)}$ if $y_i < x_i$. So ${x_i}^{(i)} \sim {y_i}^{(i)}$ in $G_i(C_i, A_i)$. This shows that G is a subgraph of $\mathcal{G}$.

As we mentioned above, it follows from [18] that $V(G) = V(\mathcal{G}) - \mathcal{V}$ so we are done with the following claim.

Claim. *Vertices x and y of G are non-adjacent in G if and only if there is some arc $(\beta+1)^{(j)} \to \alpha^{(i)}$ in A^{c} such that $\{x, y\}$ is in the set $[\alpha_{[i]}\rangle \times \langle\beta^{[j]}]$ of edges of $\mathcal{G}$.*

Proof. Let D_x and D_y be the downsets of J_L for which $e_{\mathcal{C}}(D_x) = x$ and $e_{\mathcal{C}}(D_y) = y$. Then $D_x \not\sim D_y$ if and only if there is some $(\beta+1)^{(j)} \to \alpha^{(i)}$ in A^{c} with $\alpha^{(i)}, (\beta+1)^{(j)}$ in $D_x - D_y$ (or $D_y - D_x$, but wlog we assume the former). This is true if and only if

$$y_i < \alpha \leq x_i \text{ and } y_j \leq \beta < x_j. \tag{5}$$

But since $(\beta+1)^{(j)} \to \alpha^{(i)}$ in A^{c} we certainly have that $(\beta+1)^{(j)} \leq \alpha^{(i)}$ in J_L, so x and y are not in $[\alpha_{[i]}, \beta^{[j]}]$. This means that neither of

$$(\alpha \leq x_i \text{ and } x_j \leq \beta) \text{ or } (\alpha \leq y_i \text{ and } y_j \leq \beta) \tag{6}$$

hold. As (5) and the negation of (6) are logically equivalent to

$$\alpha \leq x_i \text{ and } y_j \leq \beta,$$

we get the claim. ◇

This completes the proof of the lemma. □

Now many of the sets $[\alpha_{[i]}\rangle \times \langle\beta^{[j]}]$ in the above lemma may not actually contain edges of $\mathcal{G}$, so it begs the question: when is G an induced subgraph of $\mathcal{G}$. Clearly it is induced if and only if we can express it as $G = \mathcal{G} - \mathcal{V}$, but this does not mean that A^{c} must be empty. We have been using Lemma 4.1 to remove edges to A and assume that it satisfies (1); we may also use it to add all useless edges, doing so the complement (in J_L) is a 'reduced' version of A^{c}: a graph $\mathrm{red}(A^{\mathrm{c}})$ that generates the usual A^{c} by composition with J_L.

Corollary 5.4. *Where $\mathcal{C}$ is a chain cover of J_L, $G = G(J_L, A)$ is $\mathcal{G} - \mathcal{V}$ if and only if $\mathrm{red}(A^{\mathrm{c}})$ is a sub-digraph of $\bigcup \mathcal{C} = \bigcup C_i$.*

Proof. On the one hand, let $\mathrm{red}(A^{\mathrm{c}})$ be a subgraph of $\bigcup \mathcal{C}$. We show that no edge of $[\alpha_{[i]}\rangle \times \langle\beta^{[j]}]$, for any $(\beta+1)^{(j)} \to \alpha^{(i)}$ in $\mathrm{red}(A^{\mathrm{c}})$, is in $\mathcal{G}$. Indeed, for any $(\beta+1)^{(j)} \to \alpha^{(i)}$ in $\mathrm{red}(A^{\mathrm{c}})$ there exist k, γ and δ such that $(\beta+1)^{(j)} = (\delta+1)^{(k)}$ and $\alpha^{(i)} = \gamma^{(k)}$, so $\beta^{[j]} = \delta^{[k]}$ and $\alpha_{[i]} = \gamma_{[k]}$ and so

$$[\alpha_{[i]}\rangle \times \langle\beta^{[j]}] = [\gamma_{[k]}\rangle \times \langle\delta^{[k]}].$$

For any edge $\{x, y\}$ of $[\gamma_{[k]}\rangle \times \langle\delta^{[k]}]$ we have $\gamma \leq x$ and $y \leq \delta$, but $\{x^{(k)}, y^{(k)}\}$ is not in G_i as $(\delta+1)^{(k)} \not\to \gamma^{(k)}$ in A_k.

On the other hand, assume that $\mathrm{red}(A^{\mathrm{c}})$ is not a subgraph of $\bigcup \mathcal{C}$. Then there is some $(\beta+1)^{(j)} \to \alpha^{(i)}$ in $\mathrm{red}(A^{\mathrm{c}})$ such that $\alpha^{(i)}$ and $(\beta+1)^{(j)}$ are not both in C_k for some k. We show that the projection of $\langle\alpha^{(i)}] \not\sim \langle\beta^{(j)}]$ onto any of the G_k is an edge of G_k, and so $\langle\alpha^{(i)}] \not\sim \langle\beta^{(j)}]$ is an edge of $\mathcal{G}$. Indeed, $(\langle\alpha^{(i)}] - \alpha^{(i)}) \sim \langle\beta^{(j)}]$, or otherwise there is an arc in A^{c} in $(\langle\alpha^{(i)}] - \alpha^{(i)}) - \langle\beta^{(j)}]$ which would contradict the existence of $(\beta+1)^{(j)} \to \alpha^{(i)}$ in the reduced $\mathrm{red}(A^{\mathrm{c}})$. For any k such that $\alpha^{(i)} \notin C_k$,

$(\langle\alpha^{(i)}] - \alpha^{(i)}) \sim \langle\beta^{(j)}]$ projects onto the same edge in G_k as does $\langle\alpha^{(i)}] \not\sim \langle\beta^{(j)}]$, implying that it is in G_k. Similarily $\langle\alpha^{(i)}] \sim \langle\beta^{(j)}] \cup \{(\beta+1)^{(j)}\}$ is an arc showing that the projection of $\langle\alpha^{(i)}] \not\sim \langle\beta^{(j)}]$ onto G_k is an edge of G_k for any k such that $(\beta+1)^{(j)} \notin C_k$. □

Proof of Theorem 5.1. Let G be a reflexive DL-graph and L a compatible distributive lattice. Theorem 4.3 provides us a sub-digraph A of J_L such that $G \cong G(J_L, A)$. As we mentioned above, we have from [18] that every chain cover $\mathcal{C}$ of J_L yields an embedding $e_{\mathcal{C}} : L \cong \mathcal{P}_{\mathcal{C}} - \mathcal{V}$ into a product of chains. By Corollary 5.4 it is induced if and only if $\mathrm{red}(A^c)$ is a subgraph of $\bigcup \mathcal{C}$. We can assure this by taking every arc of $\mathrm{red}(A^c)$ as a two element chain in $\mathcal{C}$ and then covering then rest of J_L with one element chains. □

5.3. **Tight embeddings.** Theorem 5.1 tells us that for every DL-pair (G, L) there is an embedding of L into a product $\mathcal{P}$ of chains C_i such that G is an induced subgraph of the product $\mathcal{G}$ of proper interval graphs G_i. We simply refer to this as an *induced embedding* of (G, L). An embedding is *tight* if every cover of L is a cover of $\mathcal{P}$. Classical results of Birkoff and Dilworth correspond tight embeddings of L into products of chains to *chain decompositions* of J_L: chain covers $\mathcal{C}$ consisting of disjoint chains. By Lemma 5.3 any DL-pair (G, L) has a tight embedding, but if $\mathrm{red}(A^c)$ has any vertices with in-degree or out-degree greater than 2, then by 5.4 this it is not induced. In fact, we will see at the end of the next section that there are DL-graphs G such that there are no compatible lattices L for which (G, L) has a tight induced embedding. This is why we had to consider non-tight embeddings, and why we wrote [18].

6. Recognition of R-thin DL-graphs

Recall that a graph is *R-thin* if no two vertices have the same neighbourhood. As our graphs are reflexive neighbourhoods and closed neighbourhoods are the same thing.

The factorization of a categorical product was shown to be unique (up to certain obviously necessary assumptions which include R-thinness) in [7] by Dörfler and Imrich. Feigenbaum and Schäffer [9] showed that a categorical product can be factored in polynomial time. On the other hand, it was shown in [5] that proper interval graphs can be recognised in linear time. So products $\mathcal{G}$ of proper interval graphs can be recognised in polynomial time.

Using techniques discussed in [12] we will prove Theorem 1.4, which says that there is a polynomial time algorithm to decide whether or not an R-thin reflexive graph is a DL-graph. We make no effort to optimise our algorithm. First we develop some properties of DL-graphs.

6.1. **Tightest embeddings and R-thinness.** As we mentioned in Subsection 5.3, not all DL-pairs (G, L) have tight induced embeddings. We call an induced embedding *tightest* if it minimises then number of *non-tight covers*– covers of L that are not covers of $\mathcal{P}$. We prove some properties of tightest embeddings.

Claim 6.1. *If $x \prec y$ is an non-tight cover in a tightest induced embedding of a* DL-*pair (G, L), then for every vertex z of G either $z \leq x$ or $y \leq z$.*

Proof. Indeed, $x < z < y$ is impossible as $x \prec y$. If $x < z$ but $z \parallel y$ (i.e., z and y are incomparible) then $z_i > y_i$ for some i and so $x_i < (z \wedge y)_i$ giving us that $x < z \wedge y < y$, which is again impossible as $x \prec y$. Similarily $x \parallel z$ and $x < y$ is impossible. Finally, if $x \parallel z$ and $y \parallel z$ then taking $z' = x \vee z$ we get that $x < z'$ and $z' \parallel y$, which we have already seen is impossible. □

Claim 6.2. *If $x \prec y$ is an non-tight cover in a tightest induced embedding of a* DL*-pair (G, L), then for all $i \in [d]$, x_i and y_i have different neighbourhoods in G_i.*

Proof. Towards contradiction, assume that x_i and y_i have the same neighbourhoods, then replacing G_i with the proper interval graph we get by contracting x_i and y_i into a point, and reducing the i-coordinate of every vertex in $[y\rangle$ by one, we get a tighter embedding of (G, L), contradicting the fact that we started with a tigthest embedding. □

Lemma 6.3. *If G is R-thin then each G_i in a tightest induced embedding of (G, L) is R-thin.*

Proof. Towards contradiction, assume that some G_i contains vertices a and b with the same neighbourhoods. As G_i is a proper interval graph, we may assume that $b = a + 1$. By Claim 6.2 no non-tight cover projects onto $a \prec a + 1$ so there is some $x \in \mathcal{G}$ with $x_i = a$ such that x and $x + e_i$ are both in G. But these have the same neighbourhoods in $\mathcal{G}$, and so as G is an induced subgraph, they have the same neighbourhood in G, contradicting the fact that G is R-thin. □

6.2. **Removing dispensible edges to get the subgraph S.** The following definition can also be found in [12].

Definition 6.4. An edge $x \sim y$ of G *dispensable* if it satisfies the following conditions.

(i) $\exists z$ such that $N(x) \subsetneq N(z) \subsetneq N(y)$, or
(ii) $\exists z$ such that $N(y) \subsetneq N(z) \subsetneq N(x)$, or
(iii) $\exists z$ such that $N(x) \cap N(y) \subsetneq N(x) \cap N(z)$ and $N(x) \cap N(y) \subsetneq N(y) \cap N(z)$.

Observe that when G is R-thin, we can replace the $\subsetneq$ in the first two conditions with $\subset$; they are equivalent.

In the proof of the following lemma we will assume an embedding of (G, L) into some product $\mathcal{G} = \prod G_i$ of proper inteval graphs G_i. We will ust the following notation. For a vertex v_i of G_i we let $v_i^+ = \max\{N_{G_i}(v_i)\}$ be the greatest neighbour of v_i in G_i and $v_i^- = \min\{N_{G_i}(v_i)\}$ be its least neighbour. As G_i is proper interval $v_i \leq u_i$ implies that $v_i^+ \leq u_i^+$ and $v_i^- \leq u_i^-$. As G_i is R-thin, strict inequality $v_i < u_i$ implies strict inequality in at least one of $v_i^+ \leq u_i^+$ and $v_i^- \leq u_i^-$.

Lemma 6.5. *Let (G, L) be a* DL*-pair, G be R-thin, and S be the graph we get from G by removing all dispensable edges. Then*

(a) *Every edge of S is between comparable vertices of L, and*
(b) *S contains the Hasse graph $H(L)$ of L.*

Proof. First we prove part (a), by showing that any edge xy between incomparable vertices x and y is dispensable. Indeed, as x and y are incomparable, we have that $x \wedge y$ and $x \vee y$ are distinct and different from x and y. Further as $\wedge$ and $\vee$ are polymorphisms, any common neighbour of x and y is a neighbour of both of $x \wedge y$

and $x \vee y$, so $N(x) \cap N(y) \subseteq N(x \wedge y), N(x \vee y)$. By R-thinness, $N(x \wedge y)$ and $N(x \vee y)$ are distinct, so one of them properly contains $N(x) \cap N(y)$. Thus xy is dispensable.

Now, part (b) is harder. Let $x \prec y$ be a cover of L; we show that it is not dispensable. Assume some tightest induced embedding of (G, L) into a product $\mathcal{G}$ of proper interval graphs. We have two cases.

Case: $x \prec y$ is a not tight cover. For any z in $L - \{x, y\}$, we may assume by Claim 6.1 that $x \prec y < z$. We show that item i. of Definition 6.4 cannot hold. Any neighbour of y in $[y\rangle$ is a neighbour of z by (3) (of Section 2) and so any w in $N(y) - N(z)$ must be in $\langle x]$. But then by (1) it is adjacent to x, contradicting that $N(x) \subset N(z)$. Items ii. and iii. cannot hold, as by (1) any common neighbour of x and z is also a neighbour of y.

Case: $x \prec y$ is a tight cover. By the R-thinness of G, we may assume without loss of generality that there is some $v \in N(y) - N(x)$. So immediately, condition (ii) of Definition 6.4 does not hold. We may assume, by permuting indices of the interval graphs G_i, and possibly reversing the ordering on the first one, that $y_1 = x_1 + 1$

Assume that (i) holds, that is, that there is some z with $N(x) \subsetneq N(z) \subsetneq N(y)$. As z has some neighbour in $N(y) - N(x)$ we get that $x_1 < y_1 \leq z_1$, so in particular $y_1^+ < z_1^+$. As $x \sim y \sim z$ we have $z_1^- \leq y_1 \leq x_1^+$. There is some vertex w adjacent to y but neither z nor x. As it is in $N(y) - N(x)$ we have $x_1^+ < w_1 < y_1^+$. Putting these together we have

$$z_1^- \leq y_1 \leq x_1^+ < w_1 \leq y_1 < z_1^+$$

and so $w \not\sim z$ means we may assume that $w_2 \not\sim z_2$, and so that $z_2 < y_2 \leq z_2^+ < w_2$.

Now let w' be the vertex in $\mathcal{G}$ we get from x by replacing x_1 with x_1^+ and x_2 with $z_2^+ + 1$. So $w' \not\sim z$ in $\mathcal{G}$, while $w \sim x$ and $x_2 = y_2 < w_2' \leq w_2$ implies that $x_2 \sim w_2'$, so $w' \sim x$ in $\mathcal{G}$. As $N(x) \subset N(z)$ in G, w' cannot be in G, so is in some vertex interval $[\alpha_{[i]}, \beta^{[j]}]$ removed from $\mathcal{G}$ to get G.

As w is not in $[\alpha_{[i]}, \beta^{[j]}]$ we have that $j = 1$ and $x_1^+ \leq \beta < w_1$. Also, as x is not in $[\alpha_{[i]}, \beta^{[j]}]$, we have that $i = 1$ or 2. But a tightest embedding cannot have a vertex interval of the form $[\alpha_{[i]}, \beta^{[i]}]$ removed, so $i = 2$ and $x_2 < \alpha \leq w_2' = z_2^+ + 1$. So $z_2 < x_2 < \alpha \leq z_2^+ + 1$.

Now we claim that the vertex x' which we get from x by reducing x_2 to z_2 has the same neighbourhood as x in G, a contradiction. Indeed $N(x')$ contains $N(x) \cap N(z)$, so contains $N(x)$, and all vertices v of $\mathcal{G}$ that are adjacent to x but not x' have $\alpha \leq z_2^+ + 1 \leq v_2$ and $v_1 \leq x_1^+ \leq \beta$ so are in $[\alpha_{[2]}, \beta^{[1]}]$ which has been removed. Thus we have our contradiction, so (i) cannot hold.

The argument that (ii) cannot hold is essentially the same. Finally, assume that (iii) holds. Clearly this implies that both $N(x) - N(y)$ and $N(y) - N(x)$ are non-empty, so $x_1^+ < y_1^+$ and $x_1^- < y_1^-$. Moreover, z has a neighbour $a \in N(Y) - N(X)$, so having $a_1 > x_1^+$, and similarly another neighbour b having $b_1 < y_1^-$. But then there is no viable value for z_1.

This completes the proof of (b) and so of the lemma. $\square$

Compare Lemma 6.5 to similar statements in [12, Chap 8], where they show that S is closely related to what they call the *Cartesian skeleton* of a product graph G. Our proof is complicated by the fact that G is not a product, but a subgraph of a product.

6.3. **Orienting edges of S.**

Algorithm 1. *Given a graph G and subgraph S with designated vertices $\mathbf{0}$ and $\mathbf{1}$, let the sets N_j and the graphs D_j for all $j = 0, \dots, \mathrm{dist}(\mathbf{1}, \mathbf{0})$ be defined as follows. $N_j = \{v \in G \mid \mathrm{dist}(\mathbf{1}, v) = j\}$, and D_j is be the subgraph of S induced by $\cup_{\alpha=0}^{j} N_j$. Let $\vec{S}$ be the partial orientation of S we get as follows. For $j = 1, \dots, \mathrm{dist}(\mathbf{1}, \mathbf{0})$ do the following. For an edge uv of S, let $u \to v$ if*

- (i) *$u \in N_{j-1}$ and $v \in N_j$, or*
- (ii) *$u, v \in N_j$ and any one of the following holds*
  - (a) *$N(u) - N(v)$ has a vertex $u' \in D_{j-1}$ such that $u' > v'$ for all $v' \in N(v) \cap D_{j-1}$. (We consider $u' > v'$ if there is a directed path in D_{j-1} from u' to v'.)*
  - (b) *$N(v) - N(u)$ has a vertex $v' \in D_{j-1}$ such that $u' > v'$ for all $u' \in N(u) \cap D_{j-1}$.*
  - (c) *$N(v) - N(u)$ has a vertex in $N_j \cup N_{j+1}$ but not in N_{j-1}.*

If every edge of S is oriented in $\vec{S}$, and the transitive closure of $\vec{S}$ is a lattice L, then return L, otherwise, return 'NO'.

This algorithm is clearly polynomial in n.

Lemma 6.6. *Let (G, L) be a compatible pair; G be R-thin; and let S be a subgraph of G, containing the Hasse graph $H(L)$ of L, such that every edge of S is between vertices that are comparible in L. Algorithm 1, applied to S, $\mathbf{1}_L$, and $\mathbf{0}_L$, returns L.*

Proof. It is enough to show that for any (non-loop) edge uv of S with $u > v$, the above algorithm *properly orients* uv; i.e., sets $u \to v$ and at the same time does not set $v \to u$.

Observe that by construction every edge of S is either in D_j for some j or is between D_{j-1} and D_j for some j. We will prove by induction on j that the jth step of the algorithm proper orients such edges, yielding a proper orientation of all the edges of D_j. Before we do this though, we first prove that it will never improperly orient an edge.

Claim 6.7. *Let $u > v$ then the algorithm will not set $v \to u$.*

Proof. We must check that none of the conditions of the algorithm are satisfied when the roles of u and v are reversed

To see that item (i) is not satisfied observe that if not both of u and v are in N_j, then clearly it is u that is closer to 1. Indeed, if $v = x_\ell \sim x_{\ell-1} \sim \cdots \sim x_1 = 1$ is a path in G, then so is $u = u \vee x_\ell \sim u \vee x_{\ell-1} \sim \cdots \sim u \vee x_1 = 1$. So $u \in N_{j-1}$ and $v \in N_j$. (In fact this shows that the algorithm properly sets $u \to v$ in the case that u and v are not both in N_j.

To see that items (iia) and (iib) are not satisfied, it is enough to observe that if $u' \sim u$ and $v' \sim v$ and $v' \geq u'$ then $u' \sim v$ and $u \sim v'$. But this is clear, as the premises imply that

$$u' = u' \wedge v' \sim u \wedge v = v$$

and

$$v' = u' \vee v' \sim u \vee v = u.$$

To see that item (iic) is not satisfied, assume that there is some $w \in N(u) - N(v)$. As $N(u)$ is conservative, (recall the definition of conservative sets preceding Lemma 2.4) it induces a sublattice of L, so has a maximum element u'. This element must also be in $N(u) - N(v)$; as if we had $w' \sim v$, then

$$w = w \vee u' \sim u \vee v = v,$$

contradicting the fact that $w \notin N(v)$. We now show that u' is in N_{j-1}, so item (iic) is not satisfied. Indeed, some neighbour x of u must be in N_{j-1}, as $u \in N_j$. Let $x = x_{i-1} \sim x_{i-2} \sim \cdots \sim x_0 = \mathbf{1}$ be a length $i-1$ walk from x to $\mathbf{1}$. Then taking the join of each element in the walk with u' we get a walk $u' = u' \vee x_{i-1} \sim u' \vee x_{i-2} \cdots \sim u' \vee \mathbf{1} = \mathbf{1}$ from u' to $\mathbf{1}$. This shows that u' is in N_i for some $i \leq j-1$, but being a neighbour of u, it must be in N_{j-1}. $\diamond$

Now we have just to verify that for $u > v$ the algorithm sets $u \to v$.

For the case $j = 1$ let uv be an edge of S in D_1 with $u > v$. Item (i) holds if and only if $u = \mathbf{1}$, and in this case gives $u \to v$, as needed. Assume therefore that $u, v \in N_1$. As all vertices in N_1 are adjacent to $\mathbf{1}$, items (iia) and (iib) are vacuous, so we must show that (iic) holds. To see this, observe that as $u \geq v$, we have that $u_i \geq v_i$ for all $i \in [d]$. As $u_i^+ = 1 = v_i^+$ for all i, we have by R-thinness that $N_G(1) \subsetneq N_G(u) \subsetneq N_G(v)$. The vertex in $N_G(v) - N_G(u)$ is thus in N_2 as needed.

Now assume that all edges of D_{j-1} are properly oriented. We show that the jth round of the algorithm properly orients the heretofore unoriented edges of D_j.

Let uv be an edge of $D_j - D_{j-1}$ with $u \geq v$. If not both of u and v are in N_j, then (as we showed in the claim) $u \to v$ is properly ordered by step (i) of the algorithm.

So we may assume that both of u and v are in N_j. As $u > v$ we have that for all i, $u_i \geq v_i$. By R-thinness there is a vertex w in either $N(u) - N(v)$ or in $N(v) - N(u)$.

We show now that in the first case, (iia) is satisfied, and then that in the second case, (iib) or (iic) are satisfied.

Claim 6.8. *If $w \in N(u) - N(v)$, then (iia) is satisfied.*

Proof. Let $w \in N(u) - N(v)$. As we showed in the proof that item (iic) is not satisfied in the previous claim, we have that the maximum neighbour u' of u is in $D_{i-1} \cap (N(u) - N(v))$.

To see that (iia) is satisfied, we must show that $u' \geq v'$ for any neighbour v' of v in D_{i-1}. Indeed, $v' \sim v$ and $u' \sim u$ give $v' \vee u' \sim v \vee u = u$. As u' is the maximal neighbour of u this gives us that $u' \geq v' \vee u'$. This implies however that $u' = v' \vee u'$, and so $u' \geq v'$, as needed.

$\diamond$

Claim 6.9. *If $w \in N(v) - N(u)$, then (iib) or (iic) are satisfied.*

Proof. We assume that (iic) does not hold, and then show that (iib) must.

Indeed, if (iic) does not hold, then $w \in D_{j-1} \cap (N(v) - N(u))$. As $D_{j-1} \cap N(v)$ is a conservative set it induces a sublattice, so has a minimum element v'. But then for any neighbour u' of u in D_{j-1} we have from $v' \sim v$ and $u' \sim u$, that $v' \wedge u' \sim v \wedge u = v$. As D_{j-1} is conservative, $v' \wedge u'$ is in D_{j-1} so is in $D_{j-1} \cap N(v)$.

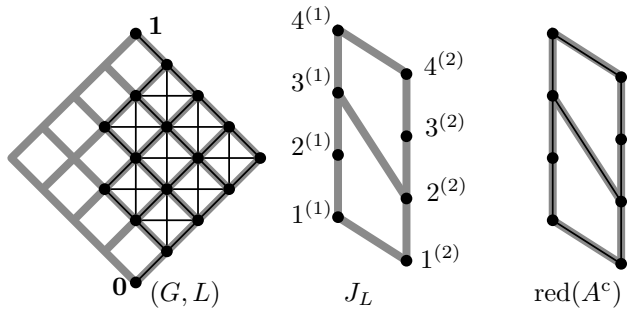


FIGURE 4. Compatible pair (G, L), poset J_L, and the graph $\mathrm{red}(A^{\mathrm{c}})$

Thus $v' \wedge u' \geq v'$ which implies that $u' \geq v'$. As $v' \notin N(u)$ we have that $u' > v'$, as needed. $\diamond$

This completes the proof of the lemma.

$\square$

Now we are ready to prove Theorem 1.4.

Proof of Theorem 1.4. Let G be an R-thin graph. It is shown in [12] that the subgraph S we get by removing dispensible edges can be found in polynomial time. For every choice of $\mathbf{0}$ and $\mathbf{1}$ in G apply Algorithm 1 to G, S and $\mathbf{0}$ and $\mathbf{1}$. As this is at most n^2 applications of a polynomial time algorithm, it is also polynomial. If for any choice of $\mathbf{0}$ and $\mathbf{1}$ a lattice L is returned then by Lemmas 6.5 and 6.6, then G is in DL and L is a compatible distributive lattice. If 'No' is returned for every choice, then by Lemmas 6.5 and 6.6, G is not in DL. $\square$

Here is an unexpected consequence of our algorithm.

Corollary 6.10. *If G is R-thin, then for a given choice of minimum and maximum vertices $\mathbf{0}$ and $\mathbf{1}$, there is at most one distributive lattice L with minimum element $\mathbf{0}$ and maximum element $\mathbf{1}$ that is compatible with G.*

With this we can get the following.

Proposition 6.11. *There are distributive lattice graphs that are not tight.*

Proof. Let (G, L) be the DL-pair shown with a tight but non-induced embedding in Figure 4. By Example 3.1, the shown $\mathbf{0}$ and $\mathbf{1}$ are the only possible $\mathbf{0}$ and $\mathbf{1}$ for lattice L compatible with G. As G is R-thin, we have by Corollary that L is the only distributive lattice (upto isomorphism) compatible with G.

The poset J_L and subdigraph $\mathrm{red}(A^{\mathrm{c}})$ are also shown. As $\mathrm{red}(A^{\mathrm{c}})$ has a vertex with up-degree two, we have, following remarks in Subsection 5.3, that there is no tight induced embedding of (G, L). So G has no compatible distributive lattice with which it has a tight induced embedding. $\square$

6.4. **Non R-thin graphs.** For a reflexive graph G we define a relation R on the vertex set by letting uRv if u and v have the same neighbourhood. Clearly this is an equivalence relations. The *R-thin reduction* of a graph G is the graph G^R whose

vertices are the sets R and in which two sets are adjacent if there are any (and so all) edges between their member vertices.

The following shows our algorithm can be useful in showing that a non R-thin graph is not DL.

Lemma 6.12. *If a reflexive graph G is a* DL*-graph then its R-thin reduction is.*

Proof. Assume that G is a reflexive DL-graph that is not R-thin. We will find pairs of vertices that are identified in G^R and show that when we identify them, or reduce the number of coordinates in which they differ, we still have a DL graph. The fact that G^R is DL then follows by induction.

For some compatible L assume a tightest induced embedding of (G, L) into $(\mathcal{G}, \mathcal{P})$. Let x and y be vertices of G with the same neighbourhood. By Lemma 6.3 there is some G_j such that x_j and y_j have the same neighbourhood in G_j. We may assume that $x_j = y_j - 1$. For any vertex v in G with $v_j \geq y_j$, reduce v_j by 1. If under this reduction, two vertices now have the same co-ordinates, then identity them- they had the same neighbourhood so are identified in G^R. Clearly there is an embedding of this reduced graph into $\mathcal{G}' = \prod G'_i$ where $G'_i = G_i$ when $i \neq j$ and we get G'_j from G_j by identifying x_j and y_j. □

We conjecture the following.

Conjecture 6.13. *For a reflexive graph G there is a polynomial time algorithm to decide whether or not G is a* DL*-graph.*

Notice that the graph G in Figure 3 is not R-thin. The vertices d and e have the same neighbourhoods. If we remove one of these vertices then the resulting lattice is distributive and is still compatible with the resulting R-thin reduction G^R. Thus the converse of the above lemma is, unfortunately, not true.

That said, one sees by reversing the operation in the proof of Lemma 6.12 that from an embedding of a DL-pair, we can add a copy of every vertex that has the same value in some coordinate. Moreover one can argue that 'fattening' the lattice in a new dimension can be replicated in the existing dimensions. So resolving the conjecture comes down solving a general version of the following game, described vaguely, but clear from Figure 5.

Given a set of numbers in a diamond tableau, decide if one can

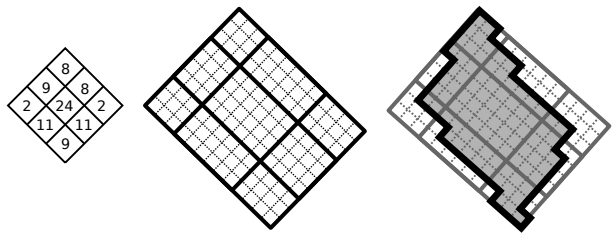
- divide the regions with square lines, and
- make two decreasing walks from the top to the bottom,

so that the number of divided regions in each of the original regions between the walks equals the number proscribed in the tableau. With some students [15], we show that this game has a polynomial time solution for tableaux of two dimentions.

7. A Question

A partial characterisation of lattice graphs can be extracted from known literature. Indeed, it follows from [13] and [16] (see also [10]) that retracts of products of reflexive paths are exactly the reflexive graphs that admit majority, or 3-NU polymorphisms, that is, polymorphisms $f : V(G)^3 \to V(G)$ satisfying

$$f(x, y, z) = c \text{ if at least two of } x, y \text{ and } z \text{ are } c.$$



Diamond Tableau Divided Tableau Two walks

FIGURE 5. The Game of Conjecture 6.13

If a reflexive graph has a compatible lattice, then it also admits the following majority operation (seen, for example, in [1])

$$f(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (x \wedge z).$$

Thus all lattice graphs are retractions of products of paths. It would also be nice to see how our characterisations can be use to show this for DL-graphs: that every DL-graph is a retract of products of paths. In general, removing a vertex interval $[\alpha_{[i]}, \beta^{[j]}]$ is not a retraction.

REFERENCES

[1] H. Bandelt, *Graphs with edge-preserving majority functions.* Discrete Math. 103 (1992) pp 1–5

[2] G. Birkhoff *Rings of sets* Duke Math. Jour. 3 (3) (1937) pp 443–454. (doi:10.1215/S0012-7094-37-00334-X).

[3] A. Bulatov, P. Jeavons, A. Krokhin, *Classifying the complexity of constraints using finite algebras.* SIAM J. Comput. 34 (2005), no. 3, 720–7

[4] C. Caravalho, V. Dalmau, A. Krokhin. *Caterpillar duality for constraint satisfaction problems* Proceed. LICS '08 (2008) pp. 307-316

[5] D. Corneil , H. Kim , S. Natarajan , S. Olariu , A. Sprague. *Simple linear time recognition of unit interval graphs* Information Processing Letters, 55 (2) (1995), pp. 99–104.

[6] R. Dilworth. *A decomposition theorem for partially ordered sets* Annals of Math. 51 (1950) pp. 161-166.

[7] W. Dorfler, W. Imrich, *Über das starke Produkt von endlichen Graphen.* Österreich. Akad. Wiss. Math.-Nature Kl.S.-B. II 18, pp.247-262.

[8] T. Feder and M.Y. Vardi. *The computational structure of monotone monadic SNP and constraint satisfaction: A study through Datalog and group theory.* SIAM Journal on Computing, 28, (1998), pp. 57–104.

[9] J. Feigenbaum, A.A.Schäffer, *Finding the prime factors of strong direct product graphs in polynomial time*

[10] T. Feder, P. Hell, B. Larose, C. Loten, M. Siggers, C. Tardif, *Graphs admitting k-NU operations. Part 1: The Reflexive Case.* SIAM J. Discrete Math. 27(4), (2013) pp. 1639-2166.

[11] F. Gardi, *A note on the Roberts characterization of proper and unit interval graphs.* Discrete Math. 307, (2007), pp. 2906–2908.

[12] R. Hammack, W. Imrich, S. Klaczer, *Handbook of product graphs.* CRC Press (2nd Edition 2011).

[13] P. Hell *Rétractions de graphes* Ph.D. Thesis, Université de Montéal (1972).
[14] P. Hell, M. Siggers, *Semilattice Polymorphisms and Chordal Graphs* Euro. Journ. of Comb. 36 (2014), pp 694–706.
[15] D.Y. Hong, S.J. Pi, M. Siggers, *A solution to the two-dimensional lattice blow-up game* Manuscript.
[16] E. Jawhari, M. Pouzet, D. Misane, *Retracts: graphs and ordered sets from the metric point of view* Combinatorics and ordered sets (Arcata, Calif., 1985), 175–226, Contemp. Math., **57**, Amer. Math. Soc., Providence, RI, 1986.
[17] I. Rival, *Maximal sublattices of finite distributive lattices. II.* Proc. Amer. Math. Soc. 44 (1974) pp. 263-268.
[18] M. Siggers, *On the representation of finite distributive lattices* Submitted (Nov. 2014) `arXiv:1412.0011 [math.CO]`.

Kyungpook National University, Republic of Korea
E-mail address: `mhsiggers@knu.ac.kr`